A conjecture of Elgin, Reading and Stella on rank-two scattering coefficients

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Abstract

For positive integers b and c, the rank-two cluster scattering diagram $\mathfrak{D}(1+x^b, 1+y^c)$ is the consistent scattering diagram with initial walls $1+x^b$ and $1+y^c$; its wall functions have coefficients $\tau^{b,c}(i,j)$. In 2024 Elgin, Reading and Stella stated eighteen conjectures on these coefficients, supported by extensive computation. Burcroff, Lee, Mou, Musiker and Reineke (2025) proved most of them and recorded Conjecture 13 as unsolved and mysterious from both the quiver-moduli and tight-grading points of view. We prove it, for the coefficients as Burcroff et al. define them, which is the convention of the table and code of Elgin, Reading and Stella: for every $j \geq 1$, $\tau^{1,5}(2j,j) = \frac{1}{j}\sum_{\ell} \binom{\ell}{j-\ell+1}\binom{j+\ell-1}{\ell}$. After a change of lattice and a monomial substitution, the wall function becomes the diagonal factor of the fifth power of an explicit automorphism of $\mathbb{Q}[\![u,w]\!]$. A grading by the difference of the two degrees and the Poisson bracket turn its factorization into a first-order differential equation whose unique solution is algebraic. The proof is formalized in Lean from the definitions.

1 Introduction

Scattering diagrams were introduced by Kontsevich and Soibelman [12] and by Gross and Siebert [11], and became a central tool for cluster algebras through the work of Gross, Hacking, Keel and Kontsevich [9]. In rank two a scattering diagram is a collection of rays in the plane, each carrying a power series called its wall function, subject to a consistency condition (Section 2). For positive integers b, c let $\mathfrak{D}(1+x^b, 1+y^c)$ be the consistent diagram whose two initial walls, on the coordinate axes, carry $1+x^b$ and $1+y^c$; it is the rank-two cluster scattering diagram associated with the cluster algebra $\mathcal{A}(b,c)$ [9, 5]. Following [5, (2.2)], for coprime $d, e \geq 1$ the coefficients $\tau^{b,c}(kd,ke)$ are defined by expanding the wall function on the ray of direction (db, ec) as $1 + \sum_{k\geq 1} \tau^{b,c}(kd,ke)\, x^{kdb} y^{kec}$.

Elgin, Reading and Stella [6] computed these coefficients symbolically to high degree and stated eighteen conjectures about them (first posted on arXiv on April 17, 2024, with Conjecture 13 already in that version; published in 2026). Their Conjecture 13 concerns $(b,c) = (1,5)$ and the wall of direction $(2,5)$. Let

$$S(t) = 1 + \sum_{j\geq 1} \tau^{1,5}(2j,j)\, t^j \in \mathbb{Q}[\![t]\!],$$

so that the wall function of $\mathfrak{D}(1+x, 1+y^5)$ on the ray of direction $(2,5)$ is $S(x^2y^5)$.

The proofs, the Lean formalization and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction.

Theorem 1.1. *For every $j \geq 1$,*

$$\tau^{1,5}(2j, j) = \frac{1}{j} \sum_{\ell=0}^{\infty} \binom{\ell}{j-\ell+1} \binom{j+\ell-1}{\ell}, \tag{1.1}$$

where only the terms with $\lceil (j+1)/2 \rceil \leq \ell \leq j+1$ are nonzero. Equivalently, $S = (1+\xi)/(1-\xi-\xi^2)$, where $\xi \in t\mathbb{Q}[\![t]\!]$ is the unique power series with $\xi = t + \xi^2 + \xi^3$; and S is the unique power series with constant term 1 satisfying

$$t^2 S^3 - 2tS^2 + (1+t)S - (1+t) = 0. \tag{1.2}$$

The first values are $\tau^{1,5}(2j, j) = 2, 5, 17, 64, 259, 1098, 4815$ for $j = 1, \ldots, 7$. Theorem 1.1 uses the convention for τ of [5], which is the convention of the table and the code of [6]; Remark 2.5 explains this point.

The right side of (1.1) was known as a sequence before the conjecture. Vladimir Kruchinin's OEIS entry A217596 (March 19, 2013), the expansion of $x/\mathrm{reversion}(x - x^2 - x^3)$, lists exactly this binomial sum, so that the right side of (1.1) is $-\mathrm{A217596}(j+1)$ [14]; in our notation its generating function is $1 - tS_*$, with S_* the series of Theorem 1.1. The same day Paul D. Hanna added functional equations for this generating function A, the first being $A = 1 - t/A - t^2/A^2$; clearing denominators gives the cubic $A^3 = A^2 - tA - t^2$, which Burcroff, Lee, Mou, Musiker and Reineke used to note that the conjectured wall function is algebraic [5, §9.3], and which is equivalent to (1.2) under $A = 1 - tS$. According to a comment of Muhammed Sefa Saydam on the entry (January 27, 2025), $-\mathrm{A217596}(n)$ counts the dissections of a convex $(n+2)$-gon into triangles and quadrilaterals by noncrossing diagonals none of which passes through a chosen vertex; so $\tau^{1,5}(2j, j)$ is this count for a $(j+3)$-gon (for example 2 for a square and 5 for a pentagon). We do not use this interpretation. What was open is that the scattering coefficients equal these numbers; this is what Theorem 1.1 proves.

Prior work

The conjectures, the data behind them and the code that produced the data [7] are due to Thomas Elgin, Nathan Reading and Salvatore Stella [6]. Amanda Burcroff, Kyungyong Lee, Lang Mou, Gregg Musiker and Markus Reineke [5] (first posted on arXiv on November 18, 2025; published in 2026) proved Conjectures 1–3, 5–10 and 14–18 of [6], combining the change-of-lattice relation between $\mathfrak{D}(1+x^b, 1+y^c)$ and $\mathfrak{D}((1+x)^c, (1+y)^b)$ (from [9, Example 1.15]), Euler characteristics of quiver moduli through the refined GW/Kronecker correspondence of Reineke and Weist [15], and the tight gradings of Burcroff, Lee and Mou [4]. They wrote that Conjectures 4, 12 and 13 remain unsolved, and that the formula of Conjecture 13 "is mysterious from both the quiver moduli and tight grading points of view" [5, §1.3 of arXiv version 1]. Ryota Akagi [1] (first posted on arXiv on July 29, 2025) proved Conjectures 1, 2, 3, 5, 6, 11 and 15 (the proof of 11 there is due to Burcroff, Lee and Mou), with results related to 14 and 16–18, by translating a description of rank-two cluster scattering diagrams through dilogarithm elements into power series, building on his earlier explicit forms of rank-two cluster scattering diagrams in low degrees [2] (first posted on arXiv on September 27, 2023). Amanda Burcroff and Kyungyong Lee [3] study broken lines and compatible pairs for rank-two quantum cluster algebras. The ordered-product factorization on which everything rests is due to Kontsevich and Soibelman [12, 13]; we use the formulation of Gross, Pandharipande and Siebert [10, §1].

What is new

The coefficients of a single wall are usually computed either by solving the consistency equations degree by degree, which involves all walls at once (this is how the data of [6] were produced), or through closed formulas for invariants attached to that wall, such as the Euler characteristics and tight gradings used in [5]. For the ray of direction $(2,5)$ the first route gives only finitely many coefficients, and, as recalled above, the authors of [5] found no explanation of the conjectured formula by the second. Here we isolate this one wall. After the change of lattice (Section 5) and the substitution $u = x$, $w = xy$ (Section 6), its fifth power becomes the diagonal factor F in the ordered factorization of T^5, where $T = \theta_{(1,0),1+u} \circ \theta_{(0,1),1+w}$, a composition of two walls (defined in (2.1)), is an explicit automorphism of $\mathbb{Q}[\![u,w]\!]$ that fixes $s = u + w$ and multiplies u/w by $1 + s$. The new step, and the one that resisted, is to obtain a closed equation for F without knowing the other walls. A grading by $h = (\text{degree in } w) - (\text{degree in } u)$ separates the factors of slope below, at and above the diagonal; writing $g_<$ for the product of the factors below the diagonal, it shows that $P = g_<^{-1}(s)$ has only three homogeneous components, $uF(uw) + \beta(uw) + w$ (Section 7). Splitting $\log(1+P)$ by this grading determines F in terms of one root of a quadratic equation (Section 8), and the Poisson bracket then turns the factorization into a first-order differential equation, whose unique solution is algebraic (Sections 9 and 10). We reuse the Kontsevich–Soibelman factorization, the change of lattice (which we reprove in the form needed) and the pentagon identity.

Organization

Section 2 gives the definitions and the convention for τ. Section 3 outlines the proof. Section 4 proves existence and uniqueness of ordered factorizations. Sections 5 and 6 reduce the theorem to the diagonal factor of T^5. Sections 7–9 derive a differential equation for that factor, and Section 10 solves it and proves Theorem 1.1. Section 11 describes the Lean formalization.

2 Scattering diagrams and the coefficients τ

This section fixes the definitions: walls, ordered products, consistency and the coefficients $\tau^{b,c}$.

2.1 Walls and ordered products

Let z_1, z_2 be variables and $\mathcal{A} = \mathbb{Q}[\![z_1, z_2]\!]$, with maximal ideal $\mathfrak{m} = (z_1, z_2)$ and the $\mathfrak{m}$-adic topology. For $a \in \mathbb{N}^2$ write $z^a = z_1^{a_1} z_2^{a_2}$ and $|a| = a_1 + a_2$, and let $\omega(a,r) = a_1 r_2 - a_2 r_1$. A *ray* is a primitive vector $r \in \mathbb{N}^2 \setminus \{0\}$, that is, one with $\gcd(r_1, r_2) = 1$; rays are ordered by their slope $r_2/r_1 \in [0, \infty]$. Let $\Lambda = 1 + t\mathbb{Q}[\![t]\!]$ be the group of one-variable power series with constant term 1.

For a ray r and $f \in \Lambda$, the *wall* $\theta_{r,f}$ is the continuous ring automorphism of $\mathcal{A}$ with

$$\theta_{r,f}(z_1) = z_1\, f(z^r)^{r_2}, \qquad \theta_{r,f}(z_2) = z_2\, f(z^r)^{-r_1}, \qquad \text{so} \qquad \theta_{r,f}(z^a) = z^a f(z^r)^{\omega(a,r)}. \tag{2.1}$$

It is invertible, with $\theta_{r,f}^{-1} = \theta_{r,1/f}$, and walls on the same ray compose by $\theta_{r,f}\theta_{r,g} = \theta_{r,fg}$. We compose automorphisms as maps, $(\phi\psi)(\eta) = \phi(\psi(\eta))$. For example, $\theta_{(1,0),1+t}$ fixes z_1 and sends z_2 to $z_2/(1+z_1)$.

Since $f(z^r) - 1 \in \mathfrak{m}^{|r|}$, a wall satisfies $\theta_{r,f}(\eta) - \eta \in \mathfrak{m}^{|r|+1}$ for every $\eta \in \mathcal{A}$. Hence, modulo $\mathfrak{m}^N$, only the finitely many walls with $|r| < N$ act nontrivially. For a *family* $f = (f_r)$ of series $f_r \in \Lambda$, one for each ray, the *ascending product* $\prod_r^{\rightarrow} \theta_{r,f_r}$ is the automorphism of $\mathcal{A}$ that agrees, modulo $\mathfrak{m}^N$ and for every N, with the finite composition $\theta_{r^{(1)},f_{r^{(1)}}} \circ \theta_{r^{(2)},f_{r^{(2)}}} \circ \cdots$ over the rays with $|r| < N$ listed by increasing slope. Thus the ray of smallest slope is leftmost and is applied last.

2.2 Consistent diagrams

Definition 2.1. Let $P_1, P_2 \in \Lambda$. A family $f = (f_r)$ is *consistent for* $\mathfrak{D}(P_1, P_2)$ if

$$\theta_{(0,1),P_2} \circ \theta_{(1,0),P_1} = \prod_r^{\to} \theta_{r,f_r}, \tag{2.2}$$

where the product runs over all rays, including $(1,0)$ and $(0,1)$. Equivalently, for every N the two sides agree on z_1 and z_2 modulo $\mathfrak{m}^N$, the right side being the finite ascending product over $|r| < N$.

By Proposition 4.2, a consistent family exists and is unique. Its entries on the axes are $f_{(1,0)} = P_1$ and $f_{(0,1)} = P_2$: on the quotient $(z_2)/(z_2^2)$ every wall with $r_2 \geq 1$ acts as the identity and $\theta_{(1,0),f}$ acts as multiplication by $f(z_1)^{-1}$, so (2.2) forces $f_{(1,0)} = P_1$, and symmetrically $f_{(0,1)} = P_2$. Its function f_r on a ray r with $r_1, r_2 > 0$ is the wall function on that ray of the diagram $\mathfrak{D}(P_1, P_2)$ of [5, §2] (Remark 2.4).

Example 2.2. For $\mathfrak{D}(1+t, 1+t)$ the consistent family has $1+t$ on the rays $(1,0)$, $(1,1)$, $(0,1)$ and 1 elsewhere: with $\mathbf{X} = \theta_{(1,0),1+t}$, $\mathbf{Y} = \theta_{(0,1),1+t}$ and $\mathbf{Z} = \theta_{(1,1),1+t}$, the identity $\mathbf{YX} = \mathbf{XZY}$ holds, both sides sending $z_1 \mapsto z_1(1+z_2)$ and $z_2 \mapsto z_2/(1+z_1+z_1z_2)$. This is the pentagon identity.

Definition 2.3. Let $b, c \geq 1$, let f be the consistent family for $\mathfrak{D}(1+t^b, 1+t^c)$, and let $d, e \geq 1$ be coprime. If v is the ray proportional to (db, ec), then $f_v(z^v)$ is a series in $x^{db}y^{ec}$, where $x = z_1$, $y = z_2$ [5, §2], and the coefficients $\tau^{b,c}(kd, ke)$ are defined [5, (2.2)] by

$$f_v(z^v) = 1 + \sum_{k \geq 1} \tau^{b,c}(kd, ke)\, x^{kdb} y^{kec}.$$

For $(b,c) = (1,5)$ and $(d,e) = (2,1)$ the vector $(db, ec) = (2,5)$ is already a ray, so $\tau^{1,5}(2j, j)$ is the coefficient of t^j in $f_{(2,5)}(t)$, where f is the consistent family for $\mathfrak{D}(1+t, 1+t^5)$. That is, $S = f_{(2,5)}$, the series of the introduction. This is the only coefficient family used below.

Remark 2.4 (Path-ordered products and orientation). In [10, §1.3], crossing a wall supported on a line through r acts by $z^a \mapsto z^a f^{\langle n_0, a\rangle}$, where $n_0 \perp r$ is primitive with $\langle n_0, \gamma'\rangle > 0$ for the crossing path γ, and a diagram is consistent when the path-ordered product $\theta_\gamma = \theta_s \circ \cdots \circ \theta_1$ along every loop is the identity. A counterclockwise loop starting just below the positive x-axis crosses the positive x-axis, the rays $\mathbb{R}_{\geq 0} r$ with $r_1, r_2 > 0$ in order of increasing slope, and the positive y-axis, each by θ_{r,f_r}^{-1} in the notation (2.1), and then crosses the negative x- and y-axes by $\theta_{(1,0),P_1}$ and $\theta_{(0,1),P_2}$. So $\theta_\gamma = \mathrm{id}$ is the identity $\theta_{(0,1),P_2}\theta_{(1,0),P_1} = \theta_{(1,0),P_1} \circ \big(\prod_{r_1,r_2>0}^{\to} \theta_{r,f_r}\big) \circ \theta_{(0,1),P_2}$, which is (2.2). Burcroff, Lee, Mou, Musiker and Reineke draw the wall carrying f_v on $\mathbb{R}_{\leq 0} v$, in the sign convention of [11], in which z^a propagates in the direction $-a$ (see [10, §1.2]); the walls carry the same functions. As a check on the orientation, Definition 2.1 with $P_1 = P_2 = (1+t)^2$ gives $(1-t)^{-4}$ on the ray $(1,1)$ and $(1+t)^2$ on $(n+1, n)$ and $(n, n+1)$, as found in [10, §1.3].

Remark 2.5 (Reading the definition in [6]). The convention of Definition 2.3 is that of the table of $\tau^{3,2}(i,j)$, $0 \leq i, j \leq 7$, printed in [6, §2], all of whose entries it reproduces (we compared them one by one), and of the code published with [6] (the method `wall_hom` of `scatcoef.py` in [7]); it is also the convention in which [5] restate the conjectures. A reader comparing with the definition in [6, §2] should note that, read literally, it attaches the exponents b and c to the two initial walls the other way round, so that its coefficient $\tau^{b,c}(i,j)$ would be $\tau^{c,b}(i,j)$ in the convention used here. Under that reading $\tau^{3,2}(3,1)$ would be 1, whereas the table in [6] prints 0, which is its value under Definition 2.3; and Conjecture 13 would already fail at $j = 1$, since $\tau^{5,1}(2,1) = 0$. We therefore read [6] in the convention of its table and code.

3 Overview of the proof

This section outlines the argument; the details follow in the same order in Sections 4–10.

Step 1: change of lattice (Section 5). The ring homomorphism $X \mapsto x$, $Y \mapsto y^5$ carries each wall of $\mathfrak{D}_5 := \mathfrak{D}((1+t)^5, 1+t)$, on $\mathbb{Q}[\![X,Y]\!]$, to a wall of $\mathfrak{D}(1+t, 1+t^5)$. Comparing the two consistent families gives $S^5 = F$, where F is the function of $\mathfrak{D}_5$ on the ray $(2,1)$. This is a case of the change of lattice of [5, Proposition 2.2].

Step 2: a single automorphism (Section 6). On $\mathbb{Q}[\![u,w]\!]$ let $T = \theta_{(1,0),1+u} \circ \theta_{(0,1),1+w}$. The substitution $u = X$, $w = XY$ carries walls to walls, T to $\mathbf{XZ}$, and the ray $(1,1)$ to $(2,1)$. With the pentagon identity this shows that F is the function on the diagonal ray $(1,1)$ in the ascending factorization of T^5. Explicitly

$$T(u) = \frac{u(1+u+w)}{1+u}, \qquad T(w) = \frac{w}{1+u},$$

so T fixes $s = u+w$ and $T(u)/T(w) = (1+s)\,u/w$.

Step 3: three components (Section 7). Group the factorization as $T^5 = g_< \circ \theta_F \circ g_>$, the factors of slope below 1, the diagonal wall $\theta_F = \theta_{(1,1),F}$, and the factors of slope above 1. Give $u^p w^q$ the degree $h = q - p$. Walls of slope above 1 only add terms of positive h, and walls of slope below 1 only terms of negative h; the diagonal wall preserves h. Because T^5 fixes s, the series $P = g_<^{-1}(s) = \theta_F g_>(s)$ can be computed from either side, and the two computations force

$$P = uF(t) + \beta(t) + w, \qquad t = uw, \quad \beta \in t\mathbb{Q}[\![t]\!].$$

Nothing else about the infinitely many other walls is used.

Step 4: splitting the logarithm (Section 8). The ratio $\mu_\phi = (\phi(u)/u)/(\phi(w)/w)$ satisfies the cocycle rule $\mu_{\phi\psi} = \phi(\mu_\psi)\,\mu_\phi$. Applied to $g_<^{-1}T^5 = \theta_F g_>$ it gives $(1+P)^5\,\mu_{g_<^{-1}} = F^2\,\theta_F(\mu_{g_>})$. Writing $1+P = (1+u\delta)\,\rho\,(1+w/\rho)$ with $\rho, \delta \in \mathbb{Q}[\![t]\!]$ and comparing the parts of the logarithm with $h<0$, $h=0$ and $h>0$ yields $F^2 = \rho^5$ and $\mu_{g_<^{-1}} = (1+u\delta)^{-5}$.

Step 5: the string equation (Section 9). The Euler derivation $E = u\partial_u + w\partial_w$ satisfies $E(s) = s$; conjugating it by the Poisson automorphism $g_<^{-1}$ shifts it by the Hamiltonian derivation of $\log \mu_{g_<^{-1}}$. This gives $(1+u\delta)(E(P)-P) = 5\{u\delta, P\}$, and with $P = (\rho+w)(1+u\delta) - 1$ the equation collapses to a single first-order differential equation for (ρ, δ), with $\delta^2 = \rho^3$, which has exactly one solution.

Step 6: the solution (Section 10). The pair (S_*^2, S_*^3) with $S_* = (1+\xi)/(1-\xi-\xi^2)$ solves the equation, so $F = S_*^5$ and $S = S_*$; Lagrange inversion gives (1.1).

The obvious approach, computing the factorization and recognizing the coefficients, does not lead to a proof: in each degree the consistency equations couple the wall of interest to all walls of smaller degree, and many of them are nontrivial already in low degree (Figure 1). Steps 3–5 avoid them: the only information about the other walls that enters is the sign of h on each side of the diagonal.

4 Ordered factorization

This section proves the existence and uniqueness of ascending factorizations (Proposition 4.2), which make Definition 2.1 meaningful and are used in every later step.

4.1 The Poisson bracket

On $\mathcal{A} = \mathbb{Q}[\![z_1, z_2]\!]$ let

$$\{f,g\} = (z_1\partial_{z_1} f)(z_2\partial_{z_2} g) - (z_2\partial_{z_2} f)(z_1\partial_{z_1} g), \qquad \text{so} \qquad \{z^a, z^b\} = \omega(a,b)\, z^{a+b}.$$

It is a continuous biderivation, and $\{\log\eta, g\} = \{\eta, g\}/\eta$ for $\eta \in 1+\mathfrak{m}$. An automorphism ϕ of $\mathcal{A}$ is *unipotent* if $\phi(z_i) \in z_i(1+\mathfrak{m})$ for $i = 1, 2$, and *Poisson* if $\phi\{f, g\} = \{\phi f, \phi g\}$ for all f, g.

Lemma 4.1. *1. A unipotent automorphism ϕ is Poisson if and only if $\{\phi(z_1), \phi(z_2)\} = \phi(z_1)\phi(z_2)$.*

2. Every wall is a unipotent Poisson automorphism, and so is every ascending product of walls.

Proof. (1) By the chain rule, both $\phi\{f, g\}$ and $\{\phi f, \phi g\}$ equal $\sum_{i,j} \phi(\partial_i f)\phi(\partial_j g)$ times $\phi\{z_i, z_j\}$, respectively $\{\phi z_i, \phi z_j\}$, and $\{z_1, z_2\} = z_1 z_2$. (2) For $g = f(z^r)$ one has $z_i\partial_{z_i} g = r_i\, q\, g$ with $q = z^r f'(z^r)/f(z^r)$, and a direct computation gives $\{z_1 g^{r_2}, z_2 g^{-r_1}\} = z_1 g^{r_2}\, z_2 g^{-r_1}\big((1+r_1 r_2 q)(1-r_1 r_2 q) + r_1^2 r_2^2 q^2\big) = \theta_{r,f}(z_1)\,\theta_{r,f}(z_2)$, so (1) applies. Products of unipotent Poisson automorphisms, and their $\mathfrak{m}$-adic limits, are again unipotent and Poisson. □

Proposition 4.2 (Kontsevich–Soibelman)**.** *Let ϕ be a unipotent Poisson automorphism of $\mathcal{A}$. There is exactly one family $f = (f_r)$ with $\phi = \prod_r^{\rightarrow} \theta_{r,f_r}$. Moreover, any automorphism of $\mathcal{A}$ has at most one such factorization.*

Proof. We give the degree-by-degree argument of [12] and [10, §1.3]. Say that two families agree below degree d if $[t^j]f_r = [t^j]f'_r$ whenever $j|r| < d$; then the corresponding ascending products agree on z_1, z_2 modulo $\mathfrak{m}^{d+1}$. If they agree below degree d, then, since every wall is the identity to first order, the coefficients of degree d enter additively at the lowest order:

$$\prod\nolimits^{\rightarrow} \theta_{r,f_r}(z_1) - \prod\nolimits^{\rightarrow} \theta_{r,f'_r}(z_1) \equiv z_1 \sum r_2\,(c_{r,j} - c'_{r,j})\, z^{jr},$$
$$\prod\nolimits^{\rightarrow} \theta_{r,f_r}(z_2) - \prod\nolimits^{\rightarrow} \theta_{r,f'_r}(z_2) \equiv -z_2 \sum r_1\,(c_{r,j} - c'_{r,j})\, z^{jr}$$

modulo $\mathfrak{m}^{d+2}$, where the sums run over the pairs (r, j) with $j|r| = d$ and $c_{r,j} = [t^j]f_r$. Distinct pairs give distinct monomials z^{jr}, and $(r_1, r_2) \neq 0$, so this difference vanishes only if the families agree below degree $d+1$. This proves uniqueness.

For existence, suppose that $\Pi = \prod^{\rightarrow} \theta_{r,f_r}$ agrees with ϕ on z_1, z_2 modulo $\mathfrak{m}^{d+1}$, with $d \geq 1$, and write $\phi(z_1) - \Pi(z_1) \equiv z_1\alpha$, $\phi(z_2) - \Pi(z_2) \equiv z_2\alpha'$ modulo $\mathfrak{m}^{d+2}$, with α, α' homogeneous of degree d. Both ϕ and Π satisfy Lemma 4.1(1); subtracting the two identities and keeping the terms of degree $d+2$ gives $z_1 z_2(z_1\partial_{z_1}\alpha + z_2\partial_{z_2}\alpha') = 0$. So for each monomial z^k of degree d, the coefficients satisfy $k_1\alpha_k + k_2\alpha'_k = 0$, that is, $(\alpha_k, \alpha'_k) = c_k\,(r_2, -r_1)$ with $r = k/\gcd(k_1, k_2)$. Adding $c_k\, t^{\gcd(k_1,k_2)}$ to f_r for each such k changes the images of z_1, z_2 only in degrees $\geq d+1$, and removes the discrepancy in degree $d+1$. The resulting families converge coefficientwise, and their limit factors ϕ. □

By Lemma 4.1, the left side of (2.2) is unipotent and Poisson, so every $\mathfrak{D}(P_1, P_2)$ has exactly one consistent family.

5 The change of lattice

This section relates the ray $(2, 5)$ of $\mathfrak{D}(1+t, 1+t^5)$ to the ray $(2, 1)$ of $\mathfrak{D}_5 = \mathfrak{D}((1+t)^5, 1+t)$. The general statement is the change of lattice of [5, Proposition 2.2], going back to [9, Example 1.15 and Appendix C]; we prove the case needed by a forward substitution.

Let $\mathfrak{D}_5$ live on $\mathbb{Q}[\![X, Y]\!]$ and $\mathfrak{D}(1+t, 1+t^5)$ on $\mathbb{Q}[\![x, y]\!]$, and let $\kappa\colon \mathbb{Q}[\![X, Y]\!] \to \mathbb{Q}[\![x, y]\!]$ be the injective continuous ring homomorphism with $\kappa(X) = x$ and $\kappa(Y) = y^5$. For a ray (d, e) put

$$v(d, e) = \begin{cases} (d, 5e) & \text{if } 5 \nmid d, \\ (d/5, e) & \text{if } 5 \mid d, \end{cases} \qquad G^{\kappa}_{(d,e)}(t) = \begin{cases} G(t)^{1/5} & \text{if } 5 \nmid d, \\ G(t^5) & \text{if } 5 \mid d, \end{cases}$$

for $G \in \Lambda$, where $G^{1/5}$ is the unique fifth root in Λ. The map $(d,e) \mapsto v(d,e)$ is a bijection of rays that multiplies slopes by 5, so it preserves their order.

Lemma 5.1. *For every ray (d,e) and $G \in \Lambda$, $\kappa \circ \theta_{(d,e),G} = \theta_{v(d,e),G^{\kappa}_{(d,e)}} \circ \kappa$.*

Proof. Both sides are continuous ring homomorphisms, so it suffices to compare them on X and Y. If $5 \nmid d$, then $v = (d, 5e)$, $z^v = \kappa(X^d Y^e)$ and $h = G^{1/5}$ give $\theta_{v,h}(x) = x\, h(z^v)^{5e} = \kappa(X\, G(X^d Y^e)^e)$ and $\theta_{v,h}(y^5) = y^5 h(z^v)^{-5d} = \kappa(Y\, G(X^d Y^e)^{-d})$. If $5 \mid d$, then $v = (d/5, e)$, $(z^v)^5 = \kappa(X^d Y^e)$ and $h(t) = G(t^5)$ give the same two values. □

Proposition 5.2. *Let G be the consistent family for $\mathfrak{D}_5$ and f the consistent family for $\mathfrak{D}(1+t, 1+t^5)$. Then $f^5_{(2,5)} = G_{(2,1)}$.*

Proof. Define a family f' on $\mathbb{Q}[\![x,y]\!]$ by $f'_{v(d,e)} = (G_{(d,e)})^{\kappa}_{(d,e)}$. By Lemma 5.1, applied to each wall, and continuity of κ, $\kappa \circ \prod^{\to} \theta_{\rho,G_\rho} = \prod^{\to} \theta_{v,f'_v} \circ \kappa$. The initial walls behave the same way: $(1,0)$ goes to $(1,0)$ with $((1+t)^5)^{1/5} = 1+t$, and $(0,1)$ goes to $(0,1)$ with $1+t^5$. Hence, with $\Phi = \theta_{(0,1),1+t^5}\theta_{(1,0),1+t}$ on $\mathbb{Q}[\![x,y]\!]$, the consistency of G gives $\Phi \circ \kappa = \prod^{\to} \theta_{v,f'_v} \circ \kappa$. So Φ and $\Pi = \prod^{\to} \theta_{v,f'_v}$ agree on x and on y^5. Both send y into $y(1+\mathfrak{m})$, and a fifth root of unity in $1+\mathfrak{m}$ is 1, so they also agree on y. Thus f' is consistent for $\mathfrak{D}(1+t, 1+t^5)$, and $f = f'$ by Proposition 4.2. In particular $f_{(2,5)} = f'_{v(2,1)} = G^{1/5}_{(2,1)}$. □

6 From $\mathfrak{D}_5$ to the fifth power of one automorphism

This section identifies $G_{(2,1)}$ with the diagonal factor of T^5 and computes T.

Let $\iota\colon \mathbb{Q}[\![u,w]\!] \to \mathbb{Q}[\![X,Y]\!]$ be the injective continuous ring homomorphism with $\iota(u) = X$ and $\iota(w) = XY$. On exponents it is the linear map $L(p,q) = (p+q, q)$, which has determinant 1, so $\omega(La, Lr) = \omega(a,r)$. It maps rays to rays lying in the cone $0 \le q \le p$, and it maps slopes by $q/p \mapsto q/(p+q)$, an increasing map; in particular $L(1,0) = (1,0)$, $L(0,1) = (1,1)$ and $L(1,1) = (2,1)$. Checking on u and w as in Lemma 5.1, one finds for every ray r and $f \in \Lambda$ that

$$\iota \circ \theta_{r,f} = \theta_{Lr,f} \circ \iota. \tag{6.1}$$

Let $\theta_u = \theta_{(1,0),1+t}$ and $\theta_w = \theta_{(0,1),1+t}$ on $\mathbb{Q}[\![u,w]\!]$, and let $T = \theta_u \circ \theta_w$. With $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$ as in Example 2.2 (on $\mathbb{Q}[\![X,Y]\!]$), (6.1) gives $\iota \circ T = \mathbf{XZ} \circ \iota$.

Proposition 6.1. *The automorphism T^5 of $\mathbb{Q}[\![u,w]\!]$ is unipotent and Poisson. Let φ be the family with $T^5 = \prod^{\to} \theta_{r,\varphi_r}$ (Proposition 4.2). Then the consistent family G for $\mathfrak{D}_5$ satisfies $G_{(2,1)} = \varphi_{(1,1)}$.*

Proof. The first sentence follows from Lemma 4.1. Let G' be the family on $\mathbb{Q}[\![X,Y]\!]$ with $G'_{Lr} = \varphi_r$ for every ray r, $G'_{(0,1)} = 1+t$, and $G' = 1$ on all other rays (those of slope in $(1,\infty)$). By (6.1), $\iota \circ T^5 = (\mathbf{XZ})^5 \circ \iota$ and $\iota \circ \prod^{\to} \theta_{r,\varphi_r} = \prod^{\to}_{\text{slope}\le 1} \theta_{\rho,G'_\rho} \circ \iota$. So $(\mathbf{XZ})^5$ and $\Pi' = \prod^{\to}_{\text{slope}\le 1} \theta_{\rho,G'_\rho}$ agree on X and on XY; since both send X to X times a unit and $\mathbb{Q}[\![X,Y]\!]$ is a domain, they agree on Y, so $(\mathbf{XZ})^5 = \Pi'$. The pentagon identity $\mathbf{YXY}^{-1} = \mathbf{XZ}$ (Example 2.2) gives $\mathbf{YX}^5 = (\mathbf{XZ})^5\mathbf{Y} = \Pi' \circ \theta_{(0,1),1+t}$. The left side is $\theta_{(0,1),1+t} \circ \theta_{(1,0),(1+t)^5}$, and the right side is the ascending product of the family G', the wall of slope ∞ being rightmost. So G' is consistent for $\mathfrak{D}_5$, $G = G'$ by Proposition 4.2, and $G_{(2,1)} = G'_{L(1,1)} = \varphi_{(1,1)}$. □

Lemma 6.2. *The automorphism T is given by*

$$T(u) = \frac{u(1+u+w)}{1+u}, \qquad T(w) = \frac{w}{1+u}.$$

Hence, with $s = u+w$, one has $T(s) = s$ and $T(u)/T(w) = (1+s)\,u/w$.

Proof. By (2.1), $\theta_w(u) = u(1+w)$, $\theta_w(w) = w$, $\theta_u(u) = u$ and $\theta_u(w) = w/(1+u)$. So $T(w) = w/(1+u)$ and $T(u) = \theta_u(u(1+w)) = u(1+\frac{w}{1+u})$. Adding the two values gives $T(s) = s$, and dividing them gives the last identity. □

From now on we write $F = \varphi_{(1,1)} \in \Lambda$, the diagonal factor of T^5. Propositions 5.2 and 6.1 give

$$S^5 = f^5_{(2,5)} = F. \tag{6.2}$$

Figure 1 shows the rays that carry nontrivial functions in the factorization of T^5 in low degree.

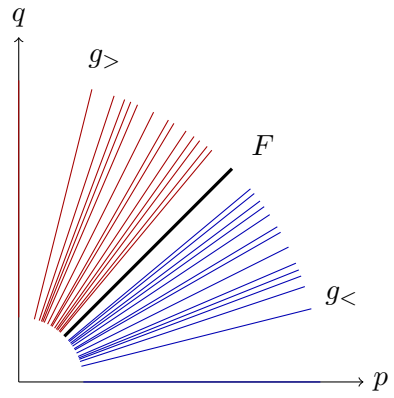


Figure 1: The rays $r = (p, q)$ whose function in the ascending factorization of T^5 has a nonzero coefficient in total degree at most 12 (that is, $[t^j]\varphi_r \neq 0$ for some $j \geq 1$ with $j|r| \leq 12$), computed by the construction in the proof of Proposition 4.2. Blue: slope below 1, the factors of $g_<$ ($h < 0$). Red: slope above 1, the factors of $g_>$ ($h > 0$). Thick black: the diagonal, whose function is F. Each ray is drawn from radius 0.9 to 4.2 for legibility. The proof uses only the sign of h on each side of the diagonal.

7 The h-grading and the extraction of the diagonal

This section introduces the grading that separates the two sides of the diagonal and shows that $g_<^{-1}(s)$ has only three graded components. Throughout, $A = \mathbb{Q}[\![u, w]\!]$ and $t = uw$; we identify $\mathbb{Q}[\![t]\!]$ with the subring of A of series in uw.

For a monomial put $h(u^p w^q) = q - p$, and for $k \in \mathbb{Z}$ let π_k be the continuous projection of A onto the closed span of the monomials with $h = k$. The monomials with $h = 0$ are the powers of t, so $\pi_0(A) = \mathbb{Q}[\![t]\!]$, and $\pi_{-1}(A) = u\mathbb{Q}[\![t]\!]$, $\pi_1(A) = w\mathbb{Q}[\![t]\!]$. Let $\mathcal{O}_{\geq k}$ and $\mathcal{O}_{\leq k}$ be the closed subspaces of series all of whose monomials have $h \geq k$, respectively $h \leq k$. Since h is additive, $\mathcal{O}_{\geq 0}$ is a closed subring, $\mathcal{O}_{\geq k}\mathcal{O}_{\geq l} \subseteq \mathcal{O}_{\geq k+l}$, and likewise for $\mathcal{O}_{\leq}$; in particular $1 + \mathcal{O}_{\geq 1}$ and $1 + \mathcal{O}_{\leq -1}$ are groups under multiplication.

Group the factorization of T^5 by slope: let $g_<$ be the ascending product of the walls θ_{r,φ_r} with $r_2 < r_1$ (including $(1, 0)$), let $g_>$ be that of the walls with $r_2 > r_1$ (including $(0, 1)$), and let $\theta_F = \theta_{(1,1),F}$, so that

$$T^5 = g_< \circ \theta_F \circ g_>, \qquad \theta_F(u) = uF(t), \quad \theta_F(w) = w/F(t). \tag{7.1}$$

The automorphisms $g_<$ and $g_>$ are Poisson, by Lemma 4.1. The following theorem, the core of the proof, is proved in Sections 7–10.

Theorem 7.1. *Let $F \in \Lambda$, let θ_F be as in (7.1), and let $g_<, g_>$ be Poisson automorphisms of $A = \mathbb{Q}[\![u, w]\!]$ such that, for every $a \in \mathbb{N}^2$,*

$$g_>(z^a) \in z^a(1 + \mathcal{O}_{\geq 1}), \qquad g_<^{-1}(z^a) \in z^a(1 + \mathcal{O}_{\leq -1}), \tag{7.2}$$

and such that $T^5 = g_< \circ \theta_F \circ g_>$. Then $F = S_*^5$, where $S_* = (1 + \xi)/(1 - \xi - \xi^2)$ and $\xi \in t\mathbb{Q}[\![t]\!]$ is the unique series with $\xi = t + \xi^2 + \xi^3$.

Lemma 7.2. *The factors in (7.1) satisfy (7.2), and θ_F commutes with every projection π_k.*

Proof. Let $\mathcal{G}_>$ be the set of unipotent automorphisms g with $g(z^a) \in z^a(1 + \mathcal{O}_{\geq 1})$ for all a. A wall with $r_2 > r_1$ lies in $\mathcal{G}_>$, because $f(z^r)^{\omega(a,r)} - 1$ is a series in the monomials z^{jr}, $j \geq 1$, which have $h = j(r_2 - r_1) \geq 1$. If $g, g' \in \mathcal{G}_>$ and $g(z^a) = z^a(1+\eta)$ with $\eta \in \mathcal{O}_{\geq 1}$, then $g'g(z^a) = g'(z^a)\,(1+g'(\eta))$, and $g'(\eta) \in \mathcal{O}_{\geq 1}$ because g' maps each monomial z^k with $h(k) \geq 1$ into $z^k(1 + \mathcal{O}_{\geq 1}) \subseteq \mathcal{O}_{\geq 1}$ and is continuous. So $\mathcal{G}_>$ is closed under composition, and it is closed under $\mathfrak{m}$-adic limits because its defining condition concerns coefficients only. Hence $g_> \in \mathcal{G}_>$. The same argument, with the inverse walls $\theta_{r,1/f}$, $r_2 < r_1$, taken in descending order, gives the statement for $g_<^{-1}$. Finally, $\theta_F(z^a) = z^a F(t)^{a_1 - a_2}$ has the same h as z^a in every term. □

Lemma 7.3. *Under the hypotheses of Theorem 7.1, the series $P = g_<^{-1}(s)$, $s = u + w$, has the form*

$$P = uF(t) + \beta(t) + w \qquad \text{for some } \beta \in t\mathbb{Q}[\![t]\!].$$

Proof. By Lemma 6.2, $T^5(s) = s$, so $P = g_<^{-1}T^5(s) = \theta_F g_>(s)$. Write $g_>(u) = u(1 + b_1)$ and $g_>(w) = w(1 + b_2)$ with $b_i \in \mathcal{O}_{\geq 1}$. Then

$$P = \theta_F g_>(s) = uF\,\theta_F(1 + b_1) + wF^{-1}\theta_F(1 + b_2),$$

where $uF\theta_F(b_1) \in \mathcal{O}_{\geq 0}$ and the last term lies in $\mathcal{O}_{\geq 1}$, by Lemma 7.2. So $\pi_{-1}(P) = uF$ and $\pi_k(P) = 0$ for $k \leq -2$. On the other side, $g_<^{-1}(u) = u(1 + \eta_1)$ and $g_<^{-1}(w) = w(1 + \eta_2)$ with $\eta_i \in \mathcal{O}_{\leq -1}$, so $P = u(1 + \eta_1) + w(1 + \eta_2)$ has $\pi_1(P) = w$ and $\pi_k(P) = 0$ for $k \geq 2$. Hence $P = uF + \pi_0(P) + w$, and $\beta = \pi_0(P) \in \mathbb{Q}[\![t]\!]$ has no constant term because $P \in \mathfrak{m}$. □

8 Splitting the logarithm

This section turns the factorization into two identities, one for F and one for $g_<$, by comparing graded components. For a unipotent automorphism ϕ of A let

$$\mu_\phi = \frac{\phi(u)/u}{\phi(w)/w} \in 1 + \mathfrak{m}.$$

Since $\phi\psi(u)/u = \phi(\psi(u)/u) \cdot \phi(u)/u$, and likewise for w, these ratios satisfy the cocycle rule $\mu_{\phi\psi} = \phi(\mu_\psi)\,\mu_\phi$.

Lemma 8.1. *Under the hypotheses of Theorem 7.1,*

$$(1 + P)^5\,\mu_{g_<^{-1}} = F^2\,\theta_F(\mu_{g_>}), \tag{8.1}$$

with $\mu_{g_<^{-1}} \in 1 + \mathcal{O}_{\leq -1}$ and $\theta_F(\mu_{g_>}) \in 1 + \mathcal{O}_{\geq 1}$.

Proof. By Lemma 6.2, $\mu_T = 1 + s$, and since T fixes s, the cocycle rule gives $\mu_{T^5} = (1+s)^5$. Also $\mu_{\theta_F} = F^2$. Apply the cocycle rule to both sides of $g_<^{-1} T^5 = \theta_F g_>$: the left side gives $g_<^{-1}((1+s)^5)\,\mu_{g_<^{-1}} = (1+P)^5 \mu_{g_<^{-1}}$, and the right side gives $\theta_F(\mu_{g_>})\,F^2$. The two memberships follow from (7.2), applied to u and w, and Lemma 7.2. □

Lemma 8.2. *Let $F \in \Lambda$ and $\beta \in t\mathbb{Q}[\![t]\!]$. There are $\rho, \delta \in \Lambda$ with $\rho\delta = F$ and $\rho + t\delta = 1 + \beta$. For such ρ, δ,*

$$1 + uF(t) + \beta(t) + w = (1 + u\delta) \cdot \rho \cdot (1 + w/\rho) \qquad \text{in } A. \tag{8.2}$$

Proof. The equation $\rho^2 - (1+\beta)\rho + tF = 0$ has a root $\rho \in \Lambda$: its coefficient of t^k is ρ_k plus a polynomial in $\rho_1, \dots, \rho_{k-1}$, so the coefficients are determined recursively. Put $\delta = F/\rho$; then $\rho + t\delta = (\rho^2 + tF)/\rho = 1 + \beta$. Finally $(1+u\delta)(\rho + w) = \rho + w + u\rho\delta + uw\delta = 1 + \beta + uF + w$. □

Proposition 8.3. *Under the hypotheses of Theorem 7.1, let β be as in Lemma 7.3 and ρ, δ as in Lemma 8.2. Then*

$$F^2 = \rho^5, \qquad \delta^2 = \rho^3, \qquad \mu_{g_<^{-1}} = (1 + u\delta)^{-5}.$$

Proof. The logarithm is a continuous isomorphism from the multiplicative group $1 + \mathfrak{m}$ onto the additive group $\mathfrak{m}$, and it maps $1 + \mathcal{O}_{\le -1}$ into $\mathcal{O}_{\le -1}$ and $1 + \mathcal{O}_{\ge 1}$ into $\mathcal{O}_{\ge 1}$, since these are closed and stable under products. Take logarithms in (8.1), using Lemma 7.3 and (8.2):

$$5\log(1+u\delta) + 5\log\rho + 5\log(1+w/\rho) + \log\mu_{g_<^{-1}} = 2\log F + \log\theta_F(\mu_{g_>}).$$

Here $u\delta \in \mathcal{O}_{\le -1}$ and $w/\rho \in \mathcal{O}_{\ge 1}$, so the terms lie in $\mathcal{O}_{\le -1}$, $\mathbb{Q}[\![t]\!]$, $\mathcal{O}_{\ge 1}$, $\mathcal{O}_{\le -1}$, $\mathbb{Q}[\![t]\!]$ and $\mathcal{O}_{\ge 1}$ respectively. Applying π_0 gives $5\log\rho = 2\log F$, that is, $F^2 = \rho^5$; then $\rho^2\delta^2 = F^2 = \rho^5$ gives $\delta^2 = \rho^3$. Applying $\sum_{k<0}\pi_k$ gives $5\log(1+u\delta) + \log\mu_{g_<^{-1}} = 0$. □

9 The string equation and a differential equation

This section uses the Poisson property of $g_<$ to obtain a differential equation for (ρ, δ), and shows that the equation has only one solution. Let $E = u\partial_u + w\partial_w$ be the Euler derivation of A. For $G \in \mathbb{Q}[\![t]\!]$ write $G' = t\,dG/dt$. One checks directly that

$$E(G) = 2G', \qquad \{G, u\} = -uG', \qquad \{G, w\} = wG', \qquad \{u, w\} = uw. \tag{9.1}$$

Lemma 9.1. *Let ϕ be a unipotent Poisson automorphism of A. Then for all $y \in A$,*

$$\phi(E(y)) = E(\phi(y)) + \{\log\mu_\phi, \phi(y)\}.$$

Proof. From the definition of the bracket, $uw\,E(y) = w\{u, y\} - u\{w, y\}$. Apply ϕ and write $\phi(u) = u\alpha$, $\phi(w) = w\alpha'$. Expanding $\{u\alpha, \phi y\}$ and $\{w\alpha', \phi y\}$ by the Leibniz rule and using the same identity for $\phi(y)$ gives $uw\alpha\alpha'\,\phi(Ey) = uw\alpha\alpha'(E(\phi y) + \{\alpha, \phi y\}/\alpha - \{\alpha', \phi y\}/\alpha')$. Divide by $uw\alpha\alpha'$, and note that $\{\alpha, g\}/\alpha - \{\alpha', g\}/\alpha' = \{\log(\alpha/\alpha'), g\}$. □

Formally, E is the Hamiltonian derivation of $\log u - \log w$, and Lemma 9.1 says that ϕ shifts $\log(u/w)$ by $\log\mu_\phi$; the lemma makes this precise without logarithms of u and w.

Proposition 9.2 (String equation)**.** *Under the hypotheses of Theorem 7.1, with P as in Lemma 7.3 and δ as in Lemma 8.2,*

$$(1 + u\delta)(E(P) - P) = 5\,\{u\delta, P\}.$$

Proof. The inverse of a Poisson automorphism is Poisson. Apply Lemma 9.1 with $\phi = g_<^{-1}$ and $y = s$. Since $E(s) = s$ and $\phi(s) = P$, and $\log \mu_\phi = -5\log(1+u\delta)$ by Proposition 8.3, it gives $P = E(P) - 5\{u\delta, P\}/(1+u\delta)$. □

Lemma 9.3. *For $\rho, \delta \in \mathbb{Q}[\![t]\!]$ put $P = (\rho + w)(1+u\delta) - 1$, and*

$$\Phi(\rho,\delta) = 1 - \rho + 2\rho' - 4t\delta - 3t\delta', \qquad \Psi(\rho,\delta) = 2\rho\delta' - 3\delta\rho'.$$

Then $(1+u\delta)(E(P) - P) - 5\{u\delta, P\} = (1+u\delta)\big(\Phi(\rho,\delta) + u\,\Psi(\rho,\delta)\big)$.

Proof. Expand the left side with the Leibniz rule and (9.1); a direct computation shows that it equals the right side. □

Corollary 9.4. *Under the hypotheses of Theorem 7.1, the series ρ and δ of Proposition 8.3 satisfy $\delta^2 = \rho^3$ and*

$$1 - \rho + 2\rho' - 4t\delta - 3t\delta' = 0. \tag{9.2}$$

Proof. By Lemma 7.3 and (8.2), $P = (\rho + w)(1+u\delta) - 1$. Proposition 9.2 and Lemma 9.3 give $(1+u\delta)(\Phi + u\Psi) = 0$. As $1 + u\delta$ is a unit, $\Phi + u\Psi = 0$; the term Φ has $h = 0$ and $u\Psi$ has $h = -1$, so $\Phi = 0$. □

Since $\delta = \rho^{3/2}$, equation (9.2) is the first-order equation $2\rho' - \rho + 1 - 3(t\rho^{3/2})' - t\rho^{3/2} = 0$ for ρ alone.

Lemma 9.5. *There is at most one pair $(\rho, \delta) \in \Lambda^2$ with $\delta^2 = \rho^3$ satisfying (9.2).*

Proof. Write $\rho = \sum \rho_k t^k$ and $\delta = \sum \delta_k t^k$, with $\rho_0 = \delta_0 = 1$. For $k \geq 1$ the coefficient of t^k in (9.2) is $(2k-1)\rho_k - (3k+1)\delta_{k-1}$. The coefficient of t^n in $\delta^2 = \rho^3$ reads $2\delta_n = 3\rho_n +$ (a polynomial in $\rho_1, \dots, \rho_{n-1}, \delta_1, \dots, \delta_{n-1}$). By induction, δ_{k-1} is determined by $\rho_1, \dots, \rho_{k-1}$, and then $\rho_k = (3k+1)\delta_{k-1}/(2k-1)$, since $2k - 1 \neq 0$. □

10 The solution and the proof of the theorem

This section exhibits the solution of (9.2), completes the proofs of Theorems 7.1 and 1.1, and derives the binomial sum.

Let $\xi \in t\mathbb{Q}[\![t]\!]$ be the unique series with $\xi(1 - \xi - \xi^2) = t$, equivalently $\xi = t + \xi^2 + \xi^3$, and let

$$S_* = \frac{1+\xi}{1-\xi-\xi^2} = \frac{1}{t} - \frac{1}{\xi}.$$

(The second expression holds because $1/\xi = (1 - \xi - \xi^2)/t$.) Since $\xi = t + O(t^2)$, the substitution $t \mapsto \xi$ identifies $\mathbb{Q}[\![t]\!]$ with $\mathbb{Q}[\![\xi]\!]$.

Lemma 10.1. *The pair $(\rho, \delta) = (S_*^2, S_*^3)$ lies in Λ^2, satisfies $\delta^2 = \rho^3$ and solves (9.2). Moreover S_* satisfies (1.2), and it is the only series in Λ that does.*

Proof. As $dt/d\xi = (1+\xi)(1-3\xi)$ is a unit, $G' = t\,(dG/d\xi)/(dt/d\xi)$ for $G \in \mathbb{Q}[\![t]\!]$, and

$$S_*' = \frac{\xi(2 + 2\xi + \xi^2)}{(1+\xi)(1-3\xi)(1-\xi-\xi^2)}.$$

With $(S_*^2)' = 2S_*S_*'$ and $(S_*^3)' = 3S_*^2S_*'$, the left side of (9.2) becomes $1 - S_*^2 + 4S_*S_*' - 4tS_*^3 - 9tS_*^2S_*'$, a rational function of ξ whose denominator is a product of the units $1+\xi$, $1-3\xi$, $1-\xi-\xi^2$; a direct computation shows that it is 0. Substituting $t = \xi(1-\xi-\xi^2)$ and S_* into (1.2) likewise gives 0. For uniqueness, the coefficient of t^k in (1.2) is S_k plus a polynomial in $S_0, \dots, S_{k-1}$. □

Proof of Theorem 7.1. By Corollary 9.4, Lemma 10.1 and Lemma 9.5, $\rho = S_*^2$ and $\delta = S_*^3$. Hence $F = \rho\delta = S_*^5$. □

Lemma 10.2. *For $j \geq 1$,*

$$[t^j]\, S_* = \frac{1}{j}\,[z^{j+1}]\,(1 - z - z^2)^{-j} = \frac{1}{j}\sum_{\ell} \binom{j+\ell-1}{\ell}\binom{\ell}{j+1-\ell}.$$

Proof. The series ξ satisfies $\xi = t\,R(\xi)$ with $R(z) = (1 - z - z^2)^{-1}$. Lagrange inversion for integer powers [8, Theorem 2.1.1 and (2.2.1)] gives $[t^n]\,\xi^k = \frac{k}{n}[z^{n-k}]R(z)^n$ for all integers k and $n \neq 0$. With $k = -1$ and $n = j$, and since $S_* = 1/t - 1/\xi$, this gives the first equality. For the second, expand $(1 - z(1+z))^{-j} = \sum_{\ell \geq 0} \binom{j+\ell-1}{\ell} z^\ell (1+z)^\ell$ and take the coefficient of z^{j+1}. □

Proof of Theorem 1.1. By Lemma 7.2, the factorization (7.1) of T^5 satisfies the hypotheses of Theorem 7.1, so $F = S_*^5$. By (6.2), $S^5 = S_*^5$. Fifth roots in Λ are unique (if $G_1, G_2 \in \Lambda$ and $G_1^5 = G_2^5$, then $G_1/G_2 \in \Lambda$ is a fifth root of unity with constant term 1, hence $G_1/G_2 = 1$), so $S = S_*$. Lemma 10.2 gives (1.1); the term with index ℓ vanishes unless $0 \leq j + 1 - \ell \leq \ell$, that is, $\lceil (j+1)/2 \rceil \leq \ell \leq j + 1$. Lemma 10.1 gives (1.2) and its uniqueness. □

Remark 10.3 (Algebraicity). In particular, the wall function $S(x^2y^5)$ on the ray $(2, 5)$ of $\mathfrak{D}(1 + x, 1 + y^5)$ is algebraic over $\mathbb{Q}(x^2y^5)$, by (1.2). Burcroff, Lee, Mou, Musiker and Reineke deduced this, and an exponential growth rate $27/5$ of the coefficients, from the conjectured formula [5, §9.3], in connection with the question, which they attribute to Kontsevich, whether every wall function of a cluster scattering diagram is algebraic. Theorem 1.1 makes both statements unconditional for this ray.

Remark 10.4 (Other exponents). Sections 6–9 do not use the value 5 except through the exponent of T. For an integer $c \geq 1$ the same argument applies to T^c and the ray $(2, 1)$ of $\mathfrak{D}((1+t)^c, 1+t)$: it shows that this ray carries $\rho_c^{c/2}$, where $\rho_c \in \Lambda$ is the unique solution of

$$2\rho' - \rho + 1 - (c-2)\big(t\rho^{c/2-1}\big)' - t\rho^{c/2-1} = 0.$$

This general form is a by-product of the written argument and is not part of the formalization, which treats $c = 5$ only; for $c = 3, 6, 7$ it agrees with the factorization of T^c through order t^5.

11 Formalization

This section states what is checked in Lean and what is trusted. The Lean development accompanying this paper (folder `lean/`) is written in Lean 4 (version 4.32.2) with Mathlib, and builds in a few minutes with the Mathlib cache.

The statement

The main theorem is `ErsConj13.ers_conjecture_13` in `lean/Main.lean`:

```
theorem ers_conjecture_13 :
    (∃ f, IsConsistentFamily (1 + PowerSeries.X) (1 + PowerSeries.X ^ 5) f) ∧
    ∀ f, IsConsistentFamily (1 + PowerSeries.X) (1 + PowerSeries.X ^ 5) f →
      ∀ j, 1 ≤ j → PowerSeries.coeff j (f ((5 / 2 : ℚ≥0) : Slope)) = ersCoeff j
```

It states that $\mathfrak{D}(1+t, 1+t^5)$ has a consistent family, so that the second part is not vacuous, and that for every consistent family f and every $j \geq 1$ the coefficient of t^j in the function on the ray of slope $5/2$, the ray $(2,5)$, equals the right side of (1.1). The definitions it uses are in `lean/ErsConj13/ScatDefs.lean` and `lean/ErsConj13/Defs.lean`:

- `Slope` is $\mathbb{Q}_{\geq 0}$ with a top element added; a slope q/p in lowest terms stands for the ray (p,q), and the top element for $(0,1)$. A family is a function from slopes to $\mathbb{Q}[\![t]\!]$.
- `wall s f` is the substitution (2.1) on $\mathbb{Q}[\![x,y]\!]$.
- `IsConsistentFamily P1 P2 f` holds when every f_r has constant term 1 and, for every N and each of x, y, the finite ascending product of the walls with $r_1 + r_2 < N$ agrees with $\theta_{(0,1),P_2}\theta_{(1,0),P_1}$ on that generator in all coefficients of total degree less than N. This is Definition 2.1, with the infinite product read degree by degree as in Section 2.
- `ersCoeff j` is $\frac{1}{j}\sum_{\ell=0}^{j+1}\binom{\ell}{j+1-\ell}\binom{j+\ell-1}{\ell}$, the right side of (1.1) (the subtractions are of natural numbers and do not truncate for $j \geq 1$, $\ell \leq j+1$), and `S` is $1 + \sum_{j\geq 1}$ `ersCoeff j` t^j.

What is proved

The Lean development assumes nothing beyond the definitions: the ordered factorization, the change of lattice and the uniqueness of the solution of (9.2) are all proved. The main intermediate results are `ErsConj13.ray21_eq_S5` in `lean/Realization.lean` ($\mathfrak{D}_5$ has a consistent family, and its function on the ray $(2,1)$ is $\mathtt{S}^5$) and `ErsConj13.main_lemma` in `lean/MainLemma.lean` (Theorem 7.1, with the conclusion $F = \mathtt{S}^5$). The table lists the Lean names of the other steps; supporting lemmas are in `lean/ErsConj13/`. Small checks of the definitions, such as the first values of `ersCoeff`, are in `lean/Examples.lean`.

In this paper	In the Lean development
Lemma 4.1	`wall_bracket`
Proposition 4.2	`factor_exists`, `factor_unique`
Proposition 5.2	`transport5_consistent`
Proposition 6.1	`transport_consistent`
Lemma 7.2	`realization`
Lemma 7.3	`V1`
Lemmas 8.1, 8.2, Proposition 8.3	`cocycle`, `quad_root`, `one_add_P_factor`, `lemma_E`
Lemma 9.1, Proposition 9.2	`euler_conj`, `lemma_SE`
Lemma 9.3, Corollary 9.4	`exact_identity`, `ode_of_SE`
Lemma 9.5	`ode_unique`
Lemma 10.1	`c5_identity`, `S_cubic`

The formal proof differs from the text in two places. It avoids logarithms: the comparison of graded components in Proposition 8.3 is replaced by the uniqueness of factorizations in $(1+\mathcal{O}_{\leq -1})\cdot\mathbb{Q}[\![t]\!]^\times\cdot(1+\mathcal{O}_{\geq 1})$ applied to (8.1) directly (`unique_factor`), and Lemma 9.1 is used in a form multiplied through by $\phi(u)\phi(w)/(uw)$. And it works with `S`, defined by the binomial sum, rather than with S_*: it proves that `S` satisfies (1.2) by showing that its coefficients and those of the root of (1.2) satisfy the same second-order linear recurrence with the same initial values (for the sequence A217596 this recurrence was recorded by Peter Bala in 2015 [14]), and then derives (9.2) for $(\mathtt{S}^2, \mathtt{S}^3)$ from (1.2). Lagrange inversion is therefore not used in the formal proof.

Axioms and trust

During the build, the file `lean/Axioms.lean` checks that `ers_conjecture_13` and the main intermediate theorems use no axioms beyond the three standard ones of Lean and Mathlib: `propext`, `Classical.choice` and `Quot.sound`. The trusted base is Lean's kernel, Mathlib, and the faithfulness of the definitions above to the coefficients $\tau^{1,5}$ of [5, (2.2)]: Definition 2.1 is the consistency condition of [10, §1.3] by Remark 2.4, the levelwise reading of the infinite product is equivalent to the $\mathfrak{m}$-adic one, and the ray of slope $5/2$ is the ray $(2,5)$ of Definition 2.3. The identification of these coefficients with those of [6] is the reading explained in Remark 2.5. As a check of conventions, separate from the proof, the values $\tau^{1,5}(2j,j)$ for $j \le 12$ produced by the code published with [6] (see [7]) agree with (1.1).

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